

The James Webb Space Telescope (JWST) is a space telescope designed primarily to conduct infrared astronomy. As the largest telescope in space, its greatly improved infrared resolution and sensitivity allow it to view objects too old, distant, or faint for the Hubble Space Telescope. This is expected to enable a broad range of investigations across the fields of astronomy and cosmology, such as observation of the first stars and the formation of the first galaxies, and detailed atmospheric characterization of potentially habitable exoplanets.

JWST was launched on 25 December 2021 on an Ariane 5 rocket from Kourou, French Guiana, and arrived at the Sun-Earth L2 Lagrange point in January 2022. The first image from JWST was released to the public via a press conference on 11 July 2022.

The telescope primary mirror consists of 18 hexagonal beryllium segments which combine to create a 6.5-meter-diameter mirror. This gives the telescope a light-collecting area about 5.6 times as large as Hubble's 2.4-meter mirror. Unlike Hubble, which observes in the near ultraviolet, visible, and near infrared spectra, JWST will observe in a lower frequency range, from long-wavelength visible light through mid-infrared.

One of the most important components of JWST is its sunshield, a five-layer diamond-shaped structure made of Kapton E, coated with aluminum and silicon. This shield protects the telescopes sensitive infrared instruments from the heat and light of the Sun, Earth, and Moon, keeping them at extremely cold temperatures necessary for operation.

The project is an international collaboration led by NASA, in partnership with the European Space Agency (ESA) and the Canadian Space Agency (CSA). Development began in 1996, but the project suffered from numerous delays and cost overruns, finally culminating in a highly successful launch and deployment.